Example 4:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed $Normal(\mu_Y, \sigma_Y^2)$ random variables.

b) Define confidence interval(s) for μ_Y as a function of $\overline{Y}$.

$\{Suppose\ \sigma_Y^2\ is\ a\ known\ constant.\}$

<u>Recall Example 6:</u> The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounce.

a) Suppose a sample of size ***n*** = 9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces? Answer the question by looking at confidence intervals for the true mean.

b) Repeat part (a) supposing that the standard deviation was 6.

c) What is the minimum sample size, ***n***, so that the sample mean will be within 0.3 ounces of the population mean, 90% of the time (upon repeated sampling)?

8.6-8.7: Large-Sample (*Z-tools*) Confidence Intervals

1. If we have an estimate such that $\hat{\theta} \approx Normal(\mu_{\hat{\theta}} = \theta, \sigma^2_{\hat{\theta}}),$ we have the following pivotal quantity,

$$Z = \frac{\hat{\theta} - \mu_{\hat{\theta}}}{\sigma_{\hat{\theta}}} = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \approx Normal(0,1).$$

2. For $Z \sim Normal(0,1),\ P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1 - \alpha.$

3. Then, $P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = P(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}} \le \theta \le \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}) = 1 - \alpha.$

   $\therefore$ The two-sided $(1-\alpha)100\%\ C.I.$ for θ is $\hat{\theta} \pm z_{\alpha/2}\sigma_{\hat{\theta}}.$

Note: If $\sigma_{\hat{\theta}}$ is unknown then use an approximated value $\hat{\sigma}_{\hat{\theta}}.$ If the sample size(s) is(are) large then $\hat{\theta} \pm z_{\alpha/2}\hat{\sigma}_{\hat{\theta}}.$ is an approximate two-sided $(1-\alpha)100\%\ C.I.$ for $\theta.$

Large-Sample (*Z-tools*) Confidence Intervals: Applications

Parameter: θ	Sample Size(s)	Estimator: $\hat{\theta}$	$\sigma^2_{\hat{\theta}}$	$\hat{\sigma}^2_{\hat{\theta}}$
μ	n	$\bar{Y}$	$\frac{\sigma^2}{n}$	$\frac{S^2}{n}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{Y}_1 - \bar{Y}_2$	$\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$	$\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$
p	n	$\hat{p}$	$\frac{p(1-p)}{n}$	$\frac{\hat{p}(1-\hat{p})}{n}$
$p_1 - p_2$	n_1 and n_2	$\hat{p}_1 - \hat{p}_2$	$\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$	$\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}$

Assumptions:

- We have large sample sizes.
  However, in the case of means, these intervals apply if we are sampling from Normal populations with known variances for any sample size(s).

- In the multiple sample applications, the samples are independent.

Example 9:

A newspaper article reports that in a poll of 120 randomly selected eligible voters, 55% said they will support candidate A in the election. There are only two candidates running. They report the results accurate within 3%.

a) Even considering the margin of error, this result indicates that candidate A will win (with between 52% and 58% support). How confident should we be in the reported margin of error?

b) If they wanted to report the results from this poll with 94% confidence, what margin of error should be reported? What is the 94% C.I.? Now comment on the results of the poll.

c) If they wanted to report the results accurate within 3%, with 98% confidence, how many observations should they have collected? Use the most conservative estimate for p.